A Quantitative Collision Principle for Odd Lipschitz Functions

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January 2026

Abstract

We study real-valued functions on metric spaces equipped with an involutive symmetry. For continuous functions that are odd with respect to the involution, path-connectedness guarantees the existence of a zero. When additional Lipschitz regularity is assumed, we prove a sharp quantitative lower bound on the distance from any point to the zero set in terms of the function value and the Lipschitz constants. The result may be viewed as a metric refinement of intermediate value–type arguments and clarifies the geometric content of symmetry-forced degeneracies.

1 Introduction

Let (V, d) be a metric space and let

$$\tau \colon V \to V$$

be an involution, i.e. $\tau \circ \tau = \mathrm{id}_V$. Functions that are *odd* with respect to such a symmetry arise naturally in many contexts, and it is well known that oddness combined with connectedness forces the existence of zeros.

The purpose of this note is twofold:

1. To isolate a minimal, domain-free existence principle for such zeros.
2. To establish a **quantitative geometric bound** describing how far any point with nonzero value must lie from the zero set, assuming Lipschitz regularity.

Importantly, the quantitative result is one-sided: Lipschitz bounds provide **lower bounds** on the distance to the zero set, but cannot guarantee proximity of a zero without additional assumptions. This distinction appears to be underemphasised in informal discussions of symmetry-based arguments.

2 Setting and Definitions

Let (V, d) be a metric space.

Definition 2.1 (Involution)**.** A map $\tau \colon V \to V$ is called an **involution** if

$$\tau(\tau(v)) = v \quad \text{for all } v \in V.$$

Definition 2.2 (Odd function)**.** A function $g\colon V \to \mathbb{R}$ is **odd with respect to** τ if

$$g(\tau(v)) = -g(v) \quad \forall\, v \in V.$$

Given a function $A\colon V \to \mathbb{R}$, define

$$g(v) := A(v) - A(\tau(v)).$$

Then g is automatically odd with respect to τ.

Definition 2.3 (Collision set)**.** The **collision set** (or zero set) associated with g is

$$Z := \{\, v \in V : g(v) = 0 \,\} = \{\, v \in V : A(v) = A(\tau(v)) \,\}.$$

3 Existence of Collision Points

We begin with a purely topological statement.

Theorem 3.1 (Existence)**.** *If V is path-connected and $g\colon V \to \mathbb{R}$ is continuous and odd with respect to an involution τ, then*

$$Z \neq \varnothing.$$

Proof. Let $v \in V$. Since V is path-connected, there exists a continuous path

$$\gamma\colon [0,1] \to V \quad \text{with} \quad \gamma(0) = v,\ \gamma(1) = \tau(v).$$

By oddness,

$$g(\gamma(0)) = g(v), \qquad g(\gamma(1)) = g(\tau(v)) = -g(v).$$

Continuity of $g \circ \gamma$ and the Intermediate Value Theorem yield $t_0 \in (0,1)$ such that $g(\gamma(t_0)) = 0$. □

Remark 3.2. No metric structure is required for Theorem 3.1. In particular, when $V = S^n$ and τ is the antipodal map, the conclusion for real-valued functions follows from oddness and connectedness alone, without invoking the Borsuk–Ulam theorem.

4 Lipschitz Regularity and Quantitative Geometry

We now impose metric regularity and study the geometry of Z.

Assumption 4.1 (Lipschitz regularity)**.** Assume:

- $A\colon V \to \mathbb{R}$ is L-Lipschitz:
$$|A(x) - A(y)| \le L\, d(x,y),$$
- $\tau\colon V \to V$ is K-Lipschitz:
$$d(\tau(x), \tau(y)) \le K\, d(x,y).$$

Lemma 4.2 (Lipschitz constant of the odd function)**.** *The function $g = A - A \circ \tau$ is Lipschitz with*

$$\mathrm{Lip}(g) \le L(1+K).$$

Proof. For any $x, y \in V$,

$$\begin{aligned} |g(x) - g(y)| &\leq |A(x) - A(y)| + |A(\tau(x)) - A(\tau(y))| \\ &\leq L\, d(x, y) + L\, d(\tau(x), \tau(y)) \\ &\leq L(1 + K)\, d(x, y). \end{aligned}$$

□

Theorem 4.3 (Distance to the collision set)**.** *For every $v \in V$,*

$$\boxed{\operatorname{dist}(v, Z) \;\geq\; \frac{|A(v) - A(\tau(v))|}{L(1 + K)}.}$$

Equivalently, for every $p \in Z$,

$$d(v, p) \geq \frac{|A(v) - A(\tau(v))|}{L(1 + K)}.$$

Proof. Since $g(p) = 0$ for all $p \in Z$,

$$|g(v)| = |g(v) - g(p)| \leq \operatorname{Lip}(g)\, d(v, p) \leq L(1 + K)\, d(v, p).$$

Taking the infimum over $p \in Z$ yields the claim. □

5 Local Consequences

Corollary 5.1 (Collision-free balls)**.** *Let $v \in V$. If g is C-Lipschitz on the ball $B(v, r)$ and*

$$r < \frac{|g(v)|}{C},$$

then

$$Z \cap B(v, r) = \varnothing.$$

Interpretation. Lipschitz regularity certifies a **minimum radius** around any point with nonzero odd value that cannot contain a collision.

6 Sharpness and Limitations

The bound in Theorem 4.3 is sharp in the sense that it cannot be reversed without additional assumptions. Lipschitz regularity acts as a **speed limit** on how fast g can decrease, but does not force a zero to occur nearby. Upper bounds on the distance to Z require extra geometric or coercivity conditions (e.g. inverse-Lipschitz behaviour along a prescribed path).

7 Conclusion

Oddness with respect to an involution guarantees the existence of zeros on path-connected spaces. When combined with Lipschitz regularity, this symmetry yields a precise quantitative statement: the magnitude of asymmetry at a point lower-bounds its distance to the symmetry-enforced zero set. The result clarifies the geometric content of symmetry arguments and may be viewed as a metric refinement of classical intermediate value phenomena.